Agile Manufacturing

Address the bottleneck in Agile hardware engineering

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SAFe & AI
SUMMIT
AMSTERDAM, NETHERLANDS

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Which aerospace company would you invest in?

	A	B
Investment:	$23B USD	$5B USD
Deliveries:	1	608

Faster production iterations have significant benefits

"The biggest problem is the SLS low flight rate, which makes every launch opportunity an experimental rather than operational procedure"

	NASA Space Launch System (SLS)	SpaceX
Estimated development costs	$23B	$5B
Number of launches	1	608 (as of Jan 2026)
Cost per launch	$2.2B	$97M (Falcon Heavy)
Max payload	95 tons	63 tons (100+ Starship)

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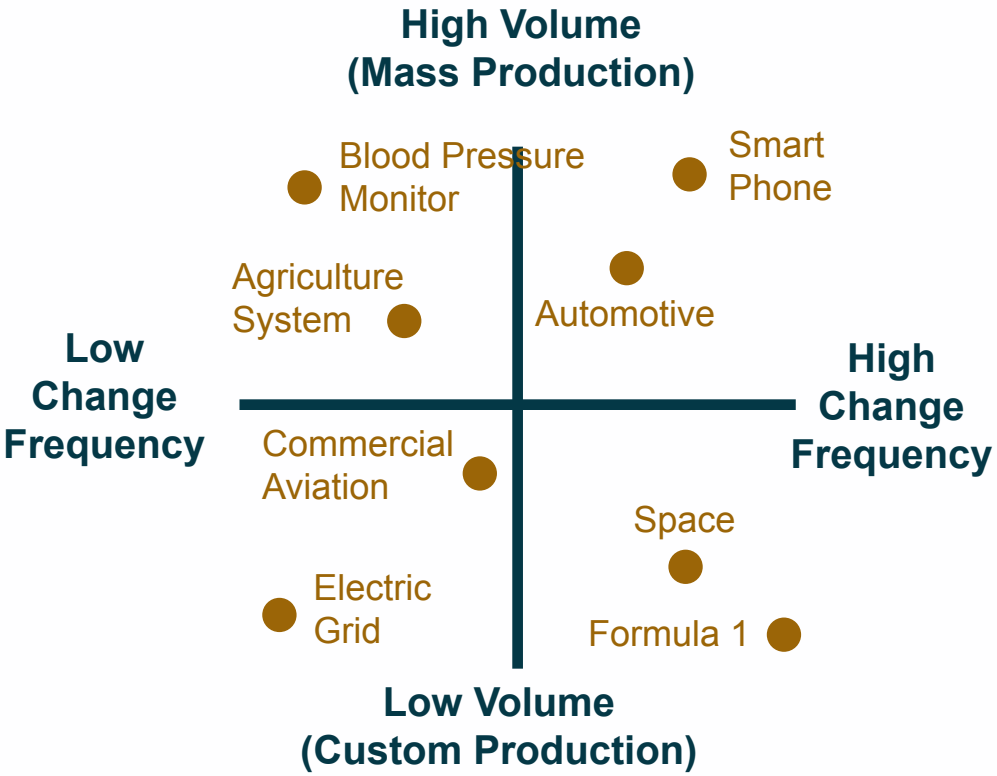
Where is your organization on the manufacturing spectrum?

- What are your volume and innovation demands?
- Where are you on the graph today?
- Where do you need to be?
- Your position helps



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Agenda

- Accelerate changes from engineering through production
- Accelerate factory changeover with digital manufacturing
- Seven practices for adopting Agile manufacturing

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Accelerate changes from engineering through production

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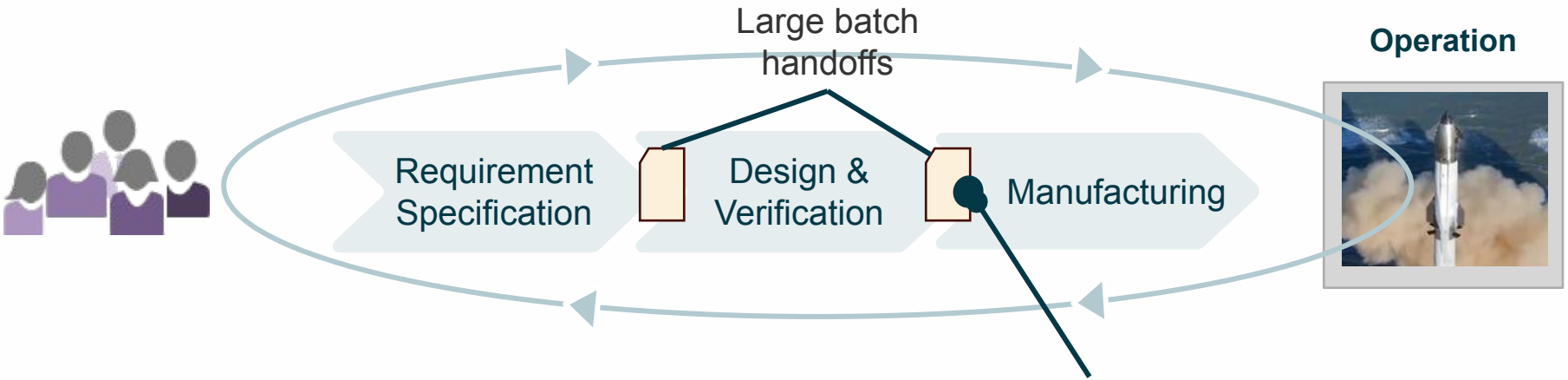
The traditional approach creates large, slow learning cycles



Factory Changeover/Start of Production

- Finalized design with complete bill of materials
- Production process flow steps
- Procedures (SOP) for operators at stations
- Acquired tooling, molds, dies, fixtures
- Robots reprogrammed and reconfigured

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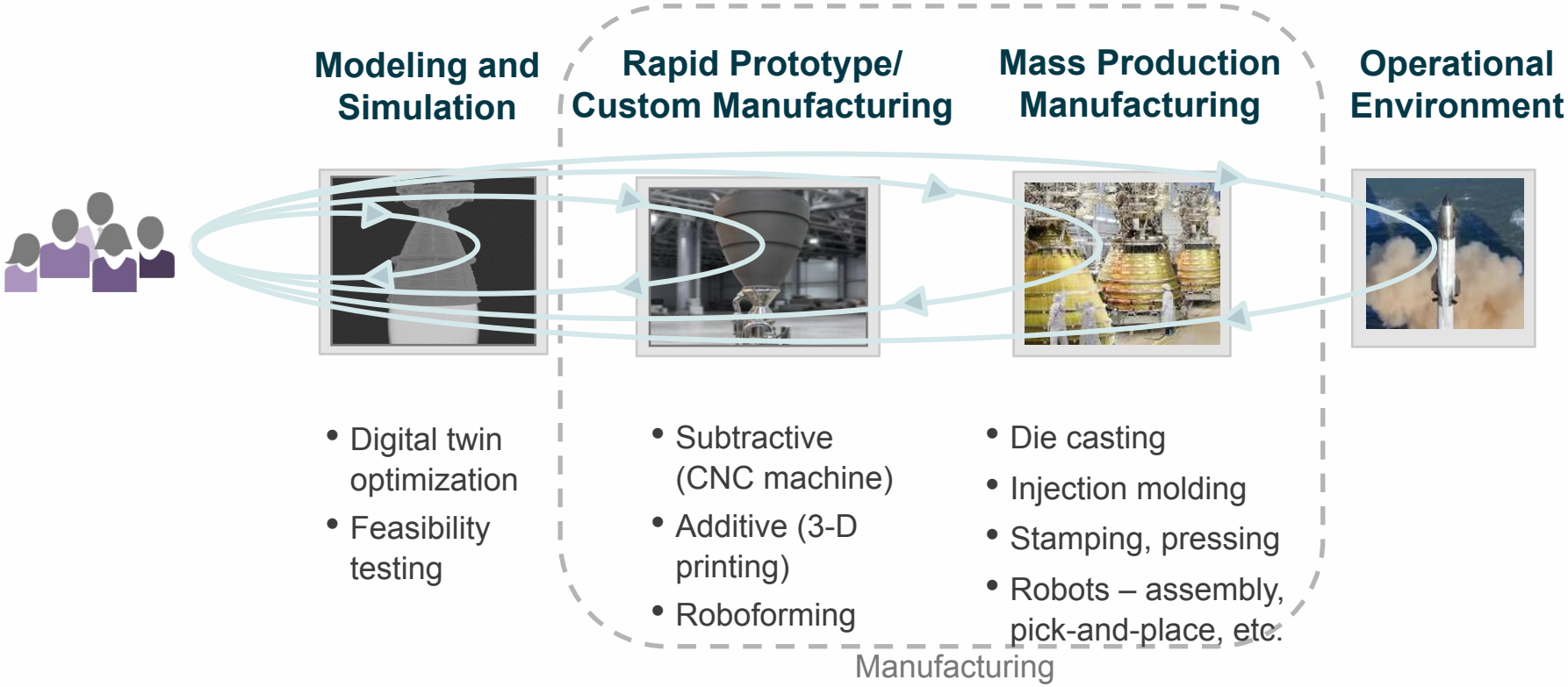
Digital technologies are accelerating learning in hardware



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The factory can be optimized for different goals

Rapid prototype/custom manufacturing

Optimized for speed of change and low up-front costs

- Slow, low-volume production
- Rapid transition from CAD design to parts
- Many manual steps

Mass production manufacturing

Optimized for production speed and volume

- Fast production, high volume
- High upfront costs and cost of change
- Mostly automated steps

Start with a factory MVP and evolve based on goals

Is the design stable (cost for design changes will increase dramatically)?

Do we have unique material requirements?

Do we have high-volume demands?

Where is the economic break-even point?

Are there quality and repeatability concerns (including certification and compliance)

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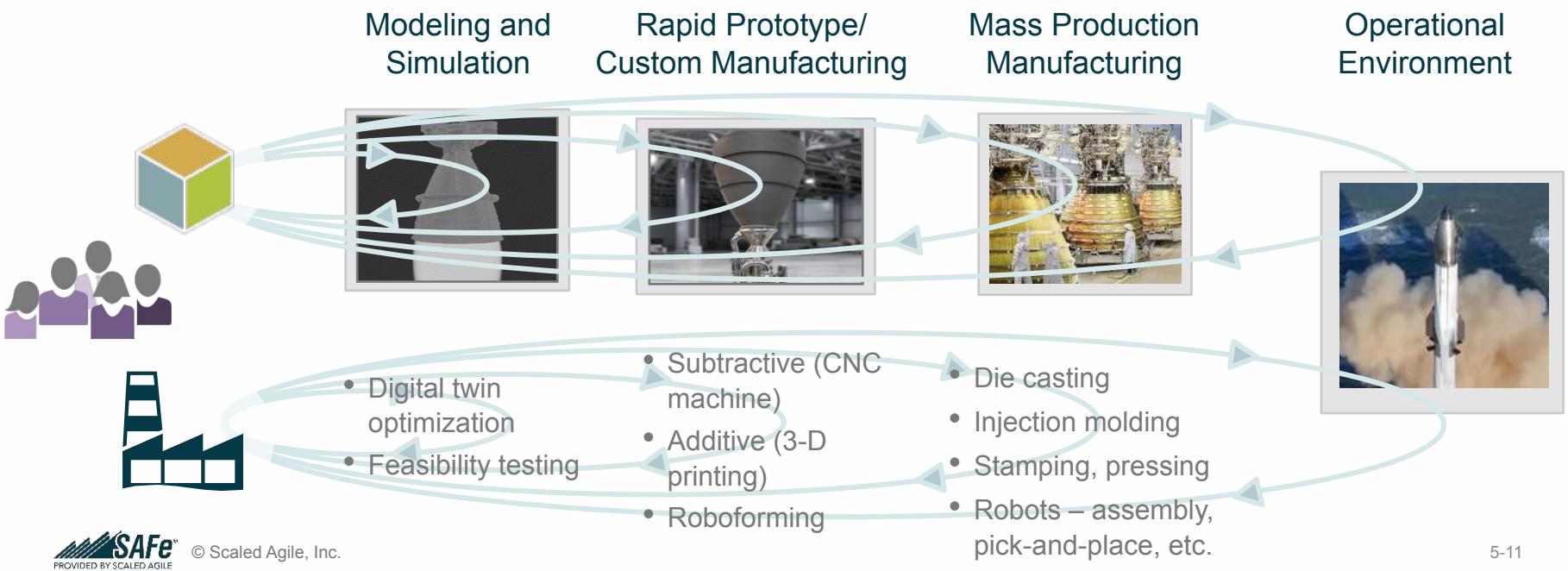
Build the product and the production system together

Evolve both based on economics as designs become stable



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Accelerate factory changeover with digital manufacturing

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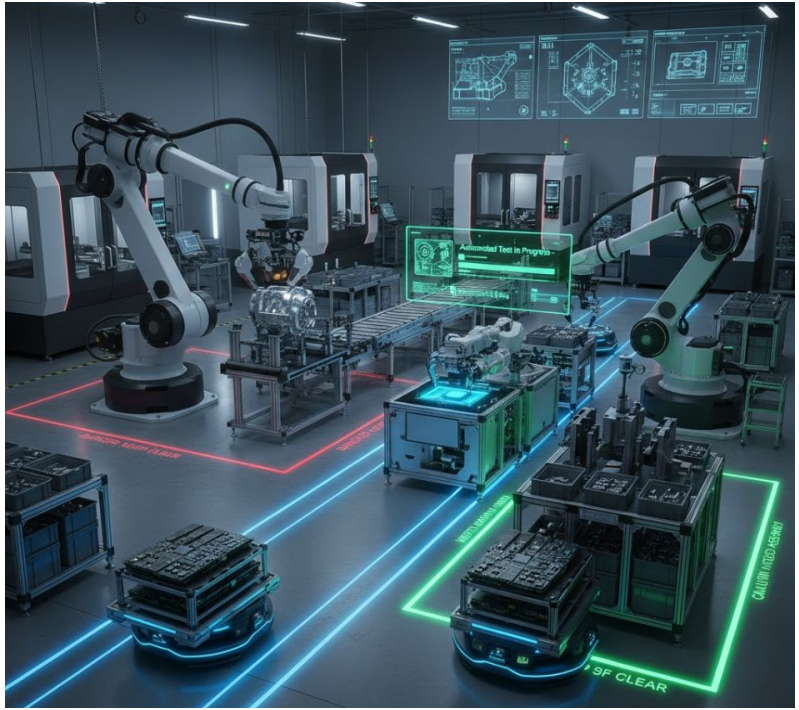
Digital manufacturing accelerates factory changeover

- Modular production cells replace a rigid line
- Sensor data in production cells instantly detects deviations, preventing defect escalation
- IoT-powered manufacturing execution system orchestrates the factory floor
- Automated guide vehicles replace conveyors
- Workers leverage digital work instructions and low-code robot re-programming
- Data collection with dashboards showing factory-wide analytics and bottlenecks



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Leverage digital technologies to build the product and the factory

Modeling and Simulation

- Digital twin of the product
- CAD, FMEA, generative designs

Rapid Prototype/ Custom Manufacturing

- Subtractive (CNC) and additive (3D printing)
- Small batch custom builds
- Hardware sprints
- *Build the product to learn, not to ship*

Digital Manufacturing

- Product designed for manufacturing at scale
- BOM finalized
- Defined process plans
- Quality gates
- Only software variants

Operations

- Product in market
- Feedback loops from field data, telemetry, and customer drive improvements (OTA)

Digital Factory

- Digital Twin of the production line
- Virtual commissioning
- Process simulation

Reconfigurable Labs & Pilot Cells

- Pilot manufacturing cells are the factory's MVP
- Small reconfigurable robot cells before committing to the entire line
- *Build the product to test the factory*

Software-Defined Factory

- Programmable platform (robots, factory floor)
- Factory OS
- Software-defined (versions, etc.)

Autonomous & Self-Optimizing Factory

- AI-powered agents make real-time decisions and manage production processes independently
- AI predictive maintenance

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True value comes from the product-factory digital integration

"A high production rate solves many ills" – Elon Musk

Modeling and Simulation	Rapid Prototype/ Custom Manufacturing	Digital Manufacturing	Operations
Digital Factory	Reconfigurable Labs & Pilot Cells	Software-Defined Factory	Autonomous & Self-Optimizing Factory

PLM Backbone: Shared Data, Backlog, and Cadence

- Design for manufacturability here, not downstream
- The product model and factory model share the same data backbone (PLM)
- PLM creates the relationship between products, processes, and resources

Build products whose primary purpose is to test the product-factory integration

- Same tool chain to change the product and factory
- A factory improvement is a software release
- Apply CI/CD to both product and production system

- Field data from the product in operations feeds back to improve the factory in operations
- E.g., warranty claims become factory changes
- Telemetry becomes design constraints

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Seven practices for adopting Agile manufacturing

1. Define modular architectures
2. Apply vertical integration within modules
3. Frequently integrate modules, virtual and physical
4. Align teams with a vision and roadmap
5. Build the factory along with the solution
6. Apply radical simplification
7. Incorporate data-driven AI management

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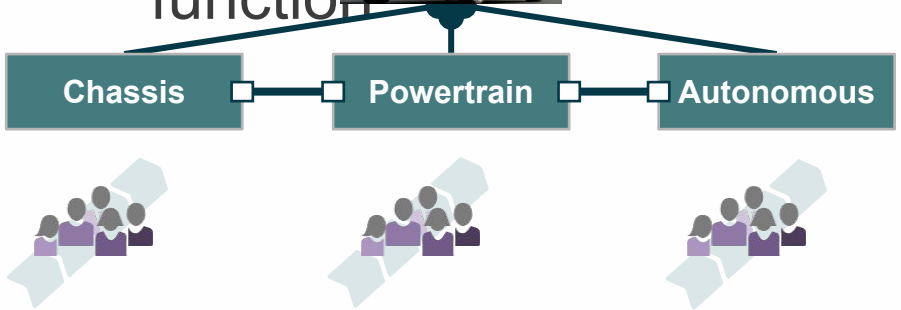
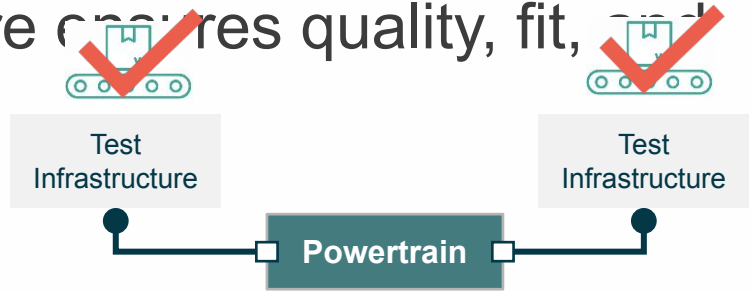
1) Define modular architectures

- Discrete modules allow teams to evolve their designs independently
- Interfaces ensure parts can be integrated and assembled quickly and reliably
- Automating infrastructure ensures quality, fit, and function





Test for characteristics:

- Physical fit
- Data/signal integrity
- Pressure, voltage, vibration, thermal, etc.

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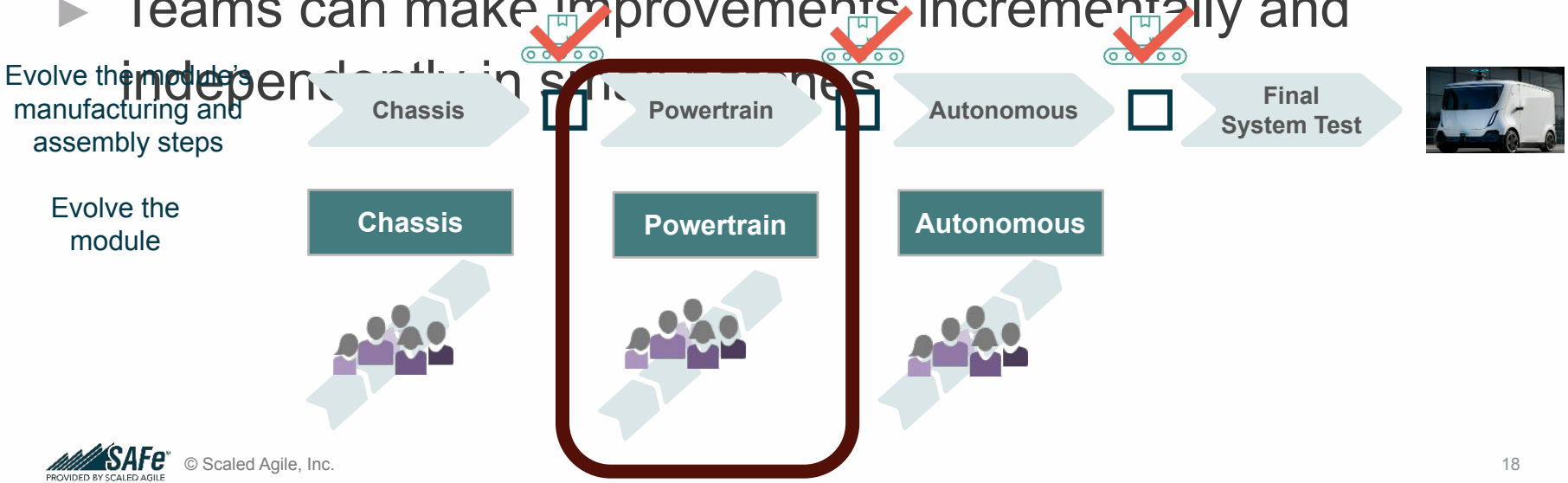
2) Apply vertical integration within modules

- Teams own the end-to-end delivery for their module
- Automated tests in the line ensure frequent changes are made reliably
- Teams can make improvements incrementally and independently in small batches



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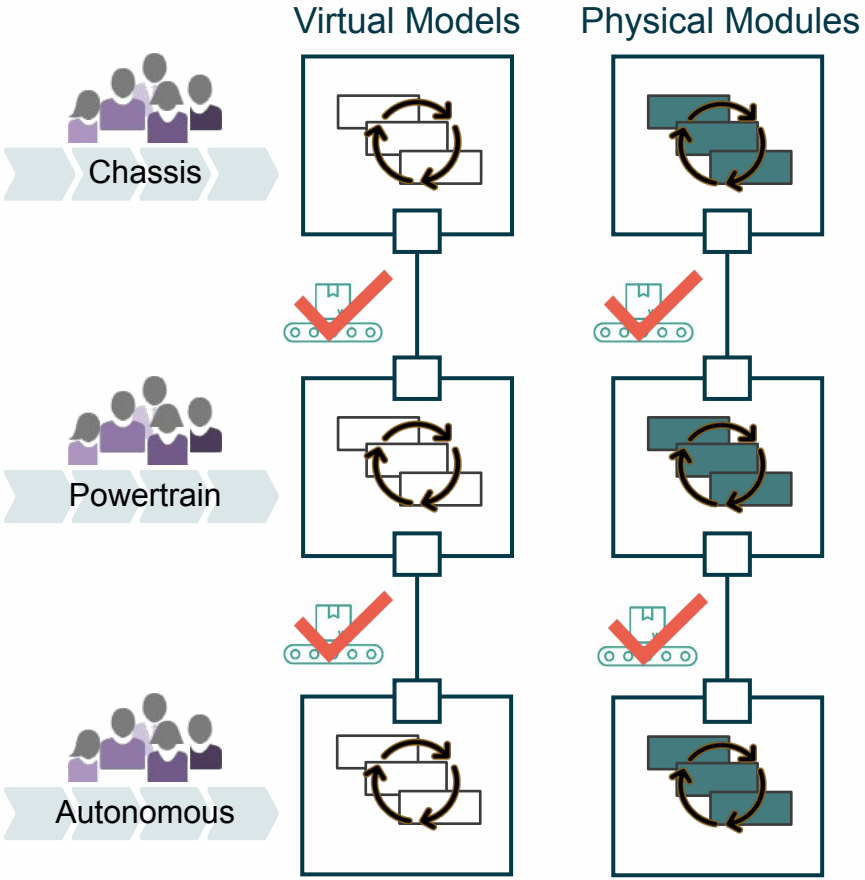
3) Frequently integrate virtual and physical modules

- Teams catch integration issues earlier for better predictability and outcomes
- Integration should occur both virtually and physically
- Automated testing ensures reliable



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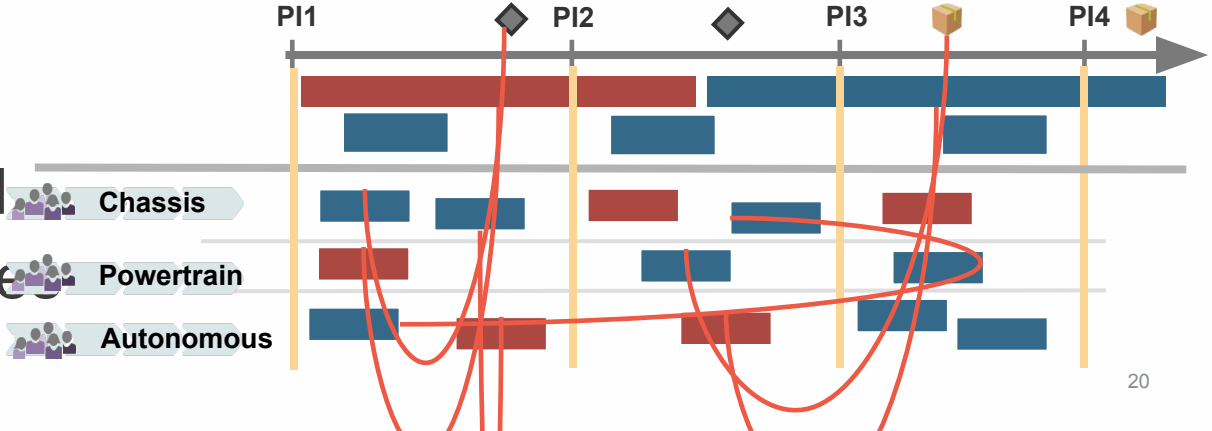
4) Align all solution builders on a shared vision and roadmap

- Inspires all solution builders to a common mission and purpose
- Creates alignment on what to build and when
- Replaces fixed specification and detailed schedules

What is our mission and goal?

What do we need to build and learn to achieve the goal?



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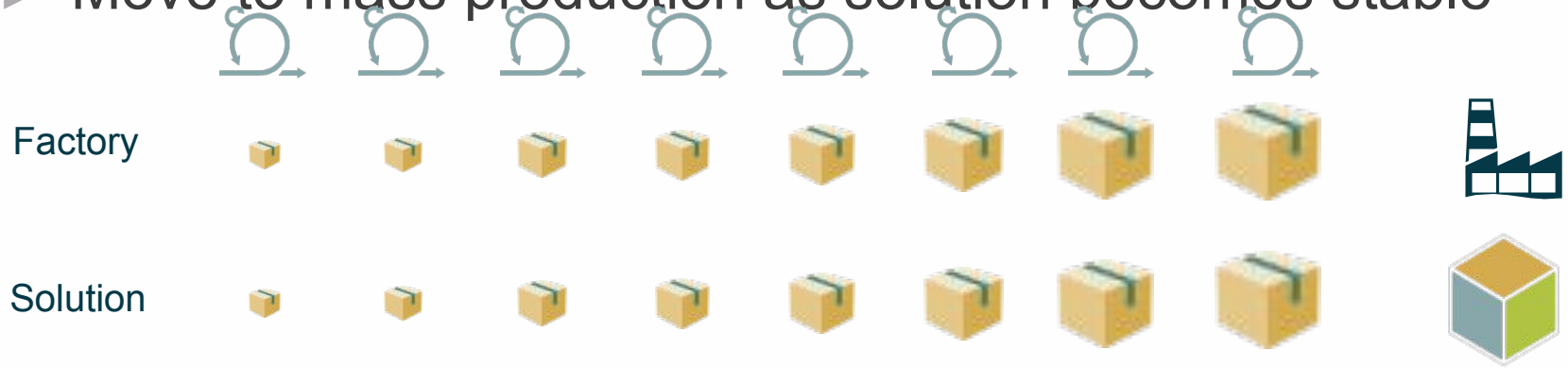
5) Build the factory along with the solution

- Hardware development must innovate in two areas:
  - The solution we deliver to customers
  - The factory that builds the solution
- Innovation in one often requires innovation in the other
- Move to mass production as solution becomes stable



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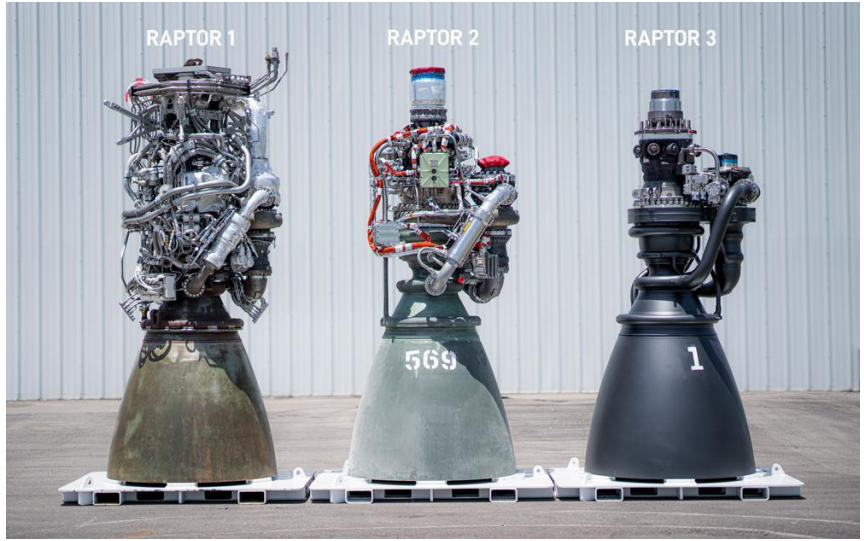
6) Apply radical simplification (“The Algorithm” from SpaceX)

1. Question every requirement
2. Aggressively remove parts or steps
3. Simplify and optimize, only after #1 and #2
4. Accelerate cycle time, only after #3
5. Automate last



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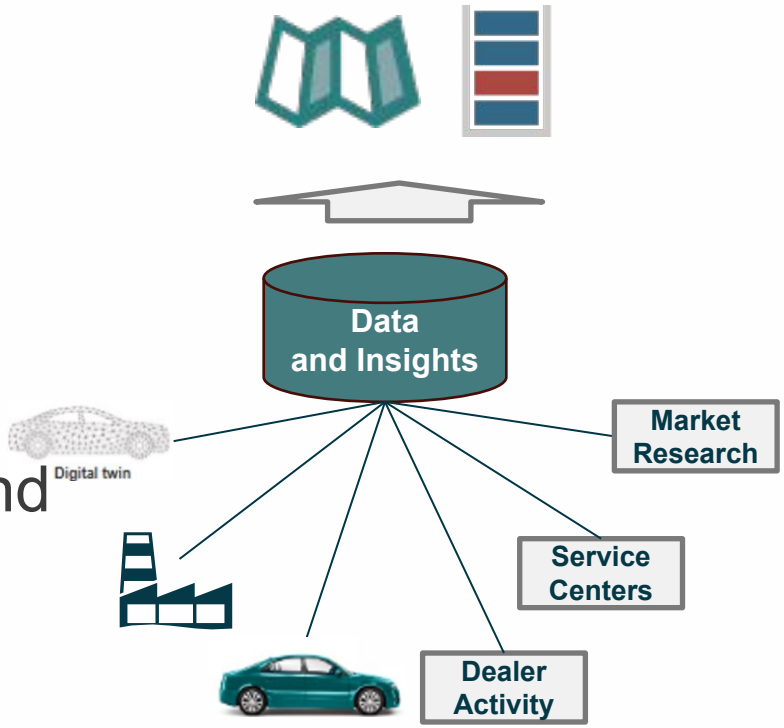
7) Incorporate data-driven AI management

- Data collection is automatic
- Data and AI replace common management decisions for prioritization, authorization, etc.
- Visible to everyone on monitors, mobile apps
- Always improving the product and the production line



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Common issues to address

- Funding
  - Fund value streams (not projects) and let them innovate and optimize
  - Invest in digitally connected factories to improve flow
- Certification and compliance
  - Move towards continuous compliance (build compliance in)
  - Certifying digital models is an emerging goal
- Suppliers
  - Integrated strategic suppliers into the Agile ways of working
  - Address procurement, contracting, and other supplier functions

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Thanks to our community members for great ideas and feedback

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Jeff Shupack
President, Project & Team

Fredrik Viljesjö
Founder, We Are Movement

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THANK YOU

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